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# Analytics for IoT-Enabled Human–Robot Hybrid Sortation: An Online Optimization Approach

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Motivated by a realworld practice of a China Post sortation center, this study considers the deployment of Internet of Things (IoT) technology to improve the efficiency of a human–robot hybrid sortation system. In this system, IoT technology enables responsive adjustment of the manual capacity to alleviate the congestion effect of increasing parcel flow on robotic sorting efficiency. We begin with a predictive analysis of a real-life dataset to quantify the congestion effect on robotic sorting efficiency, and examine the nonstationary behavior of the parcel flow process. Subsequently, we design an online algorithm using IoT advance information (which refers to advance observation of some parcel flows before their actual arrivals) for efficiently adjusting the manual capacity. We develop theoretical guarantee for the effectiveness of the online algorithm by bounding its regret, and also demonstrate that the long-term expected regret rate is zero under mild conditions. Via extensive simulation experiments, we find that our online algorithm outperforms the China Post sortation center's current policy and conventional Markov dynamic program in terms of computational efficiency and solution quality. The simulation results also demonstrate that the value of IoT technology to the sortation center can be significant, and shed insights on IoT investment by revealing the diminishing return of expanding the horizon of IoT advance information.

*Key words:* human–robot hybrid parcel sortation; manual capacity; online optimization; IoT program
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## 1. Introduction

The ongoing COVID-19 pandemic has led to a significant increase in the parcel volume in many countries, and the global parcel volume is predicted to reach 130 billion by 2021. The shipment of such a large number of parcels creates overwhelming pressure on every phase of parcel logistics, especially parcel sorting and distribution. To improve the parcel sorting efficiency, modern sortation centers have begun to deploy robotic technologies (e.g., cross-belt sorters, robotic arms, and automated guided vehicles) to process parcels (Boysen et al. 2019). It is well known that robots are more efficient than human workers at handling repetitive tasks in an accurate and accelerated manner. However, human workers outperform robots in terms of better flexibility and adaptability to volatile environments. To combine the advantages of robots and human workers, modern sortation centers extensively use human–robot hybrid systems as effective solutions for parcel sorting. A well-known example of such hybrid systems is Amazon's fulfillment centers that employ robots as assistants for human pickers.

The management of human–robot hybrid systems can be rather complicated. The key challenge is regarding how to improve the parcel sorting efficiency by enabling connection and coordination between robots and human workers. Fortunately, the Internet of Things (IoT), another emerging technology, has the potential to be an enabler of tightly connected and well-coordinated human–robot hybrid systems. Under IoT architecture, interconnected sensors are extensively installed inside/outside a sortation center. These sensors are responsible for collecting real-time data about trucks, parcel flows, robots, and human workers, and sending the data to an IoT software platform in the cloud. Fed with dynamically updated data, the IoT software platform is able to monitor the process of parcel sorting and output responsive decisions that coordinate the activities of robots and human workers (Lee et al. 2018).

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### 1.1. An Example of IoT-Enabled Human–Robot Hybrid Sortation

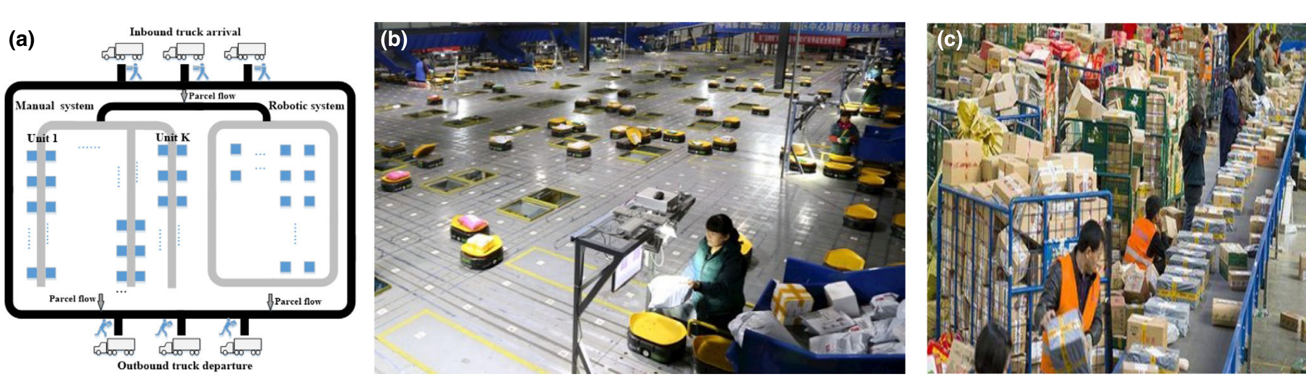
This study is partially motivated by our consulting experience with a China Post sortation center. This sortation center is operating 24/7 as follows. Everyday, a batch of inbound parcels (carried by trucks) are scheduled to arrive at the sortation center at the beginning of a predetermined equal-length time window (called "period" hereafter). The sortation center employs a human–robot hybrid system (see Figure 1a) to sort out the inbound parcels according to their destinations during the period. The central part of the hybrid system is a robotic system with more than 200 robots (see Figure 1b). When the inbound parcel flow is small or moderate, these robots can efficiently and accurately sort out the parcels. Accordingly, the robotic system alone can provide satisfactory sortation performance as its average processing time (per parcel), the sortation center's key performance indicator (KPI), is very small. However, when the parcel flow is substantial, the average processing time of the robotic system becomes large because the congestion effect of increasing parcel flow on reducing the robotic efficiency intensifies. To reduce the burden of the robotic system, a proportion of the parcel flow is shared with the manual system consisting of a set of identical sortation units (see Figure 1c).

The robotic system is always online and has high efficiency relative to the manual system under the same conditions; however, its capacity is inflexible to adjust especially when the parcel flow is large.[1] Conversely, although the manual system has low efficiency, its capacity is flexible to adjust by turning on/off some manual sortation units. The sortation center manager's key task is to reduce the average processing time of the hybrid system cost-effectively by dealing with two-dimension questions: (a) how to allocate the parcels between the robotic and manual systems? (b) How to adjust the manual capacity responsively subject to volatile parcel flows over time? Particularly, Question (b) is a challenge because of both technical and operational difficulties in adjusting the manual capacity. On the technical level, responsive manual capacity adjustment requires efficient information exchange and well coordination of some effort-consuming procedures (like, equipment preparation and workforce scheduling) in the sortation center. On the operational level, it is nontrivial to determine how many manual sortation units to turn on/off subject to highly uncertain parcel flows.

Currently, the information exchange over conventional information systems is inefficient; meanwhile, the procedures of adjusting manual capacity are less coordinated. These two disadvantages prevent the China Post sortation center from implementing the aforementioned responsive manual capacity adjustments. Instead, the sortation center has to use a *static* approach for planning the manual capacity as follows. The entire planning horizon is divided into a set of subhorizons with a few periods in each subhorizon. At the beginning of every subhorizon, a capacity plan that claims the manual capacity for every period of the subhorizon is made before the corresponding parcel flows actually arrive; during the subhorizon, the capacity plan is strictly executed. In the present, this static approach can be irresponsive to volatile parcel flows; for example, we observe that the China Post sortation center often switches on all the manual sortation units under the static approach, leading to unnecessarily high operating cost.

Recently, the China Post sortation center initializes an IoT program that enables responsive manual capacity adjustment. The implementation of the IoT program involves the renovation of the sortation center's infrastructure by installing hundreds of sensors and a cloud software. Under the IoT architecture, information about parcels, trucks, robots, and human workers is automatically collected by sensors, and is

**Figure 1** **Illustrations for the China Post Sortation Center: (a) 2D-Layout of the Human–Robot Hybrid System, (b) Snapshot of the Robotic System, and (c) Snapshot of a Manual Sortation Unit [Color figure can be viewed at wileyonlinelibrary.com]**

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timely exchanged between relevant parties via smartphones, tablets, and dashboards. Consequently, the parcel logistics information becomes more transparent in the sense that the sortation center is able to track in-transit parcels in advance to their actual arrivals. Moreover, the cloud software enables the manager to design algorithms for calculating the manual capacity decisions based on real-time flow information. All these benefits of the IoT program lay the foundation for responsive manual capacity adjustment.

### 1.2. Research Questions and Methods

The above discussion has shown that IoT technology is the potential enabler of improving the sorting efficiency of human–robot hybrid systems. However, the theoretical understanding on IoT-enabled human–robot hybrid sortation is still limited. This paper attempts to provide an analytics study on such hybrid sortation systems by exploring the following questions: (i) How to quantify the congestion effect of increasing parcel flow on the sorting efficiency of the robotic system? (ii) How can manual capacity be dynamically optimized in an IoT context? (iii) How to assess the value of the IoT program at the sortation center?

To answer these questions, we first conduct a predictive analysis of a real-life dataset, provided by the China Post sortation center, to quantify the congestion effect of increasing parcel flow on reducing the sortation efficiency of robotic system and to characterize the underlying parcel flow process. Specifically, we develop a function that characterizes the effect of potential feature variables on the average processing time of the robotic system, and estimate a time series model that captures the behavior of the parcel flow process. Based on the predictive finding, we develop a theoretical model that formulates the China Post sortation center's sequential decision problem of managing the human–robot hybrid system over a planning horizon. In every period, the manager decides the number of online manual sortation units ("manual capacity decision") and the proportions of parcel flow allocated to robotic and manual systems ("flow allocation decision") before and after the parcel flow arrives, respectively. Accordingly, the manager collects the following costs: (a) a penalty cost that is increasing in the average processing time of the hybrid system, (b) a quadratic operating cost that is increasing in the number of online manual sortation units, and (c) a setup cost for expanding manual capacity. Particularly, the manual capacity decisions play a central role in shaping the performance of the hybrid system, which are difficult to optimize because the consideration of the setup cost couples the manual capacity decisions over different periods together. Supported by IoT technology, the manager is able to dynamically update manual capacity according to the advance information about parcel flows before their actual arrivals (called *IoT advance information*). The manager's objective is to minimize the expected cost rate based on the cost-efficiency tradeoff.

Although our analysis is based on the practice of the China Post sortation center, our results are restricted neither to the China Post company nor to the region of China. In fact, human–robot hybrid systems have been widely used by a few giant retail platforms, including Amazon, Walmart, and Kmart, in many countries, such as the United States and Europe. Furthermore, "how to improve the sorting efficiency by flexibly and timely adjusting robotic/manual capacities to cope with the volatile parcel flows over times?" is a fundamental problem for all the platforms. Our work provides a systematic framework that integrates predictive and theoretical analyses to address this problem. Furthermore, the results can shed insights into an increasingly popular topic, namely the value of IoT technology in operations management (Choi et al. 2021, Sun and Ji 2021, Tang 2021). Therefore, we believe that our findings have broad applications in the domain of IoT-enabled human–robot collaboration in smart warehouses.

### 1.3. Key Findings and Contributions

We begin with a conventional linear regression analysis of the dataset and find that the congestion effect on reducing robotic efficiency is significant and mainly shaped by increasing parcel flow. Two generalized regressions—isotonic regression and shape-constrained polynomial regression—are employed to develop more accurate functions that predict the average processing time of robotic system using parcel flow as the feature variable. On one hand, the isotonic regression generates a piecewise step function, which has been used in the literature to model the congestion effect in warehousing facilities (Choi et al. 2021, Sun and Ji 2021, Tang 2021). On the other hand, the shaped-constrained polynomial regression generates a smooth function incorporating prior structural knowledge about the average processing time function (Curmei and Hall 2020). The cross-validation test shows the predictive performance of the two generalized regressions is comparable and better than that of the conventional linear regression. We also find that the underlying parcel flow process follows a non-stationary log-ARIMA(1,1,1)-without-drift process. The primary implications of the predictive analysis are as follows. First, although the piecewise step function and shape-constrained polynomial function are both good fit to the data, we chose the polynomial function to model the average processing time of the robotic system in the theoretical analysis because of

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its good structural properties (e.g., differentiability). Second, the non-stationary nature of the parcel flow motivates us to develop algorithms that use IoT advance information to dynamically update the manual capacity decisions.

In the theoretical analysis, we first consider the flow allocation problem and derive the optimal flow allocation decisions based on the convexity of the objective function. Next, we consider the manual capacity adjustment problem and focus on developing an online algorithm, referred to as the "variant online gradient descent" (VOGD), to solve the capacity problem in an online fashion. The VOGD makes full use of IoT advance information to generate the capacity decisions using the gradients of the cost functions. Moreover, given a sample path of parcel flows, we demonstrate that the regret of our VOGD is upper bounded by a term that is convexly decreasing as the horizon of IoT advance information expands. It is noteworthy that the manual capacity problem can be solved using the conventional Markov dynamic program (MDP) as well, under the assumption that the underlying parcel flow process follows the non-stationary log-ARIMA(1,1,1)-without-drift process. However, our online algorithm has several distinctive features compared to the MDP approach. First, the computational efficiency of the VOGD is high, but that of the high-dimensional MDP can be low without useful structural properties to reduce the search space. In this sense, the VOGD is more compatible with IoT architecture under which decisions have to be made within a very short timeframe (even a few seconds). Second, the VOGD deals with a more challenging version of the manual capacity adjustment problem without prior distributional knowledge about the underlying parcel flow process. Third, to our knowledge, it is difficult to develop a guarantee for the effectiveness of the MDP approach. Instead, we demonstrate that under mild conditions, the long-term expected regret rate of the VOGD is zero, which provides a statistical guarantee for the effectiveness of our online algorithm.

We further validate the effectiveness of our online algorithm relative to the static approach and the MDP approach via extensive simulation experiments. We find that the VOGD outperforms the two benchmark approaches in terms of computational efficiency and solution quality. Specifically, the VOGD generates robust and near-optimal capacity decisions that result in an average performance loss of 0.22% (relative to the optimal cost in hindsight) within 1 second; while, the MDP approach generates suboptimal capacity decisions that lead to an average performance loss of 20.13% within 3330 seconds. We also assess the value of the IoT program to the China Post sortation center, that is, the cost reduction brought by the online algorithm relative to the current static approach. The average cost reduction is 97.86%, implying that the value of the IoT program can be significant.

Our study makes several contributions to the literature. First, this is one of the few analytics studies that explore the congestion effect of increasing parcel flow on the robotic sortation efficiency based on a real-life dataset. Our finding extends the literature by introducing the shape-constrained polynomial function as another way to model the congestion effect in warehousing facilities. Second, we develop an online algorithm that fully utilizes IoT advance information to update the manual capacity decisions, together with theoretical guarantee on the effectiveness of our online algorithm. To the best of our knowledge, our work is the first to design an online algorithm for solving the parcel sortation problem in the IoT context. Third, the extensive simulation experiments based on the real-life dataset provide insights on the adoption IoT technology in human–robot hybrid systems. From the numerical results, we observe a significant diminishing return brought by expanding the horizon $w$ of IoT advance information to increase the value of the IoT program. This observation reminds decision-makers to carefully choose $w$ because the expansion in the horizon of IoT advance information is costly, however, it may not generate sufficient surplus to cover the investment.

## 2. Literature Review

This study is related to several areas of OM research on (i) parcel sorting, (ii) human–robot hybrid warehousing, and (iii) capacity management subject to demand uncertainties.

*Parcel Sorting*. This stream of literature addresses the problem on how to improve the parcel sorting efficiency, from different perspectives such as layout design (Fedtke and Boysen 2017, Petersen 2000), inbound/outbound truck scheduling (Boysen et al. 2017), and destination to sorter (or to outbound-door) assignment (Jarrah et al. 2016, Novoa et al. 2018). Similar to these papers, our work focuses on reducing average processing time of a human–robot hybrid sortation system. However, our paper has some distinctive features. First, few of the existing literature empirically/theoretically considers the congestion effect in parcel sorting; while, our paper provides a research framework that integrates both empirical and theoretical analyses to study the congestion effect. Second, the majority of the papers studies parcel sorting problems in the presence of stationary parcel flows, and develops offline algorithms for solving large-scale mixed-integer program problems (Khir et al. 2021, Novoa et al. 2018). Differently, our work is carried out under non-stationary environments, and

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focus on developing online algorithms. Third, many papers consider the management of a single type of sorting system (either automated or manual system); an exception is the work of Khir et al. (2021), which considers the serial combination of an automated and a manual system, and optimizes the dispatch schedule between the two systems. Unlike Khir et al. (2021), our paper considers the parallel combination of an automated and a manual system, and explores the adjustment of manual capacity and the allocation of parcel flow between these two systems.

*Human–Robot Hybrid Smart Warehousing*. A growing number of OM papers considers order picking problems in human–robot hybrid warehouses. For example, Wang et al. (2021) studies robot scheduling problem that takes into account the schedule-induced fluctuation of the working states of order pickers. Löffler et al. (2021) and Azadeh et al. (2020), respectively, study picker routing and dynamic zoning of picking area in a robot-supported picking system; while Pasparakis et al. (2021) examines productivity of different types of robot-supported picking system (i.e., human-following and human-leading systems) through field experiments. Our work differs from the literature by studying a parcel sorting problem, and to our knowledge, it is the first to examine the issue of flow allocation and capacity adjustment in the human–robot hybrid warehouse. Furthermore, our predictive finding (i.e., the shape-constrained polynomial function can be used to model the congestion effect in warehouses) is new to the literature, and extends the classic papers (e.g., Zhang et al. 2009) that often use piecewise linear functions to model the congestion effect in warehouse facilities. Broadly speaking, this study is also relevant to these papers that study the design and application of IoT-based smart warehouses (Lee et al. 2018), and is one of the few analytics studies that investigate the operational value of IoT technology in smart facilities.

*Capacity Management subject to Demand Uncertainties*. This stream of literature studies several topics such as capacity investment, capacity allocation, and capacity flexibility. Some papers address the issues of capacity investment by analyzing a two-stage model in which the capacity investment (capacity allocation) decision is made before (after) demand realizations (Netessine et al. 2002, Notz and Pibernik 2021). Several other papers study the problem of allocating limited and fixed capacity/supply to sequentially arrived demands using forecast or advance information (Kim et al. 2021, Papier 2016). Differently, the capacity in our paper is flexible to adjust (by turning on/off some manual sortation units), and one of our focus is on how to efficiently conduct this adjustment. The other researchers are interested in examining the value of capacity flexibility under the assumption that decision makers can update their capacity decisions (Bish and Wang 2004, Qi et al. 2017). Among these papers, our paper is closely relevant to the work of Qi et al. (2017), which focuses on a firm's optimal capacity adjustment decision with demand learning over a stationary demand arrival process. Unlike their work, our paper considers a non-stationary parcel arrival process. Furthermore, their approach is based on the conventional theory of stochastic dynamic programming, which requires the prior knowledge on the demand pattern or distribution family about the demand. While, our approach is based on the theory of online convex optimization, which does not need such prior knowledge.

From the perspective of online convex optimization, our work is close to Li et al. (2021), who studies the online algorithm for problems with precise prediction and quadratic switching costs. Unlike their work that assumes every single-period cost function to be strongly convex and sufficiently smooth (called $l$-smoothness), our paper only requires the strong-convexity, and thus generalizes their work by relaxing a key assumption.

## 3. Predictive Analysis

The predictive analysis is based on a raw dataset offered by the China Post sortation center. This dataset describes the operations of the human–robot hybrid system over a time horizon of 3 months (from February to April 2020). The first part of the dataset records the information about all the parcels processed by the hybrid system, including the fields such as parcel ID, parcel weight, arrival time, and destination. The second part describes the sorting efficiency of the robotic system, based on which the processing time of every parcel (seconds) can be computed. Utilizing this dataset, we attempt to quantify (a) the congestion effect of increasing parcel flow on the sorting efficiency of the robotic system, and examine (b) the underlying parcel flow process. It is worth mentioning that the China Post sortation center does not have the data about the operations of its manual system because of technical reasons. We turn to ask the sortation center's managers to provide an approximate estimate about the sorting efficiency of the manual system based on their experience.

The raw dataset is preprocessed as follows. First, according to the time schedule of truck arrivals, we divide the entire time horizon into a sequence of consecutive periods. The length of every single period is 3 hours. A parcel flow arrives at the sortation center at the beginning of every period, and it is completely processed at the end of the period. Second, we count the number $d_t$ of the parcel flow arriving at the sortation center in period $t(=1, 2, \ldots)$. We also count the number $\hat{d}_t(\leq d_t)$ of the parcel flow processed by the robotic system in period $t$. Third, we calculate the average processing time $p_t$ and average weight $w_t$ of the parcels processed by the robotic system in period $t$. Subsequently, we obtain another dataset with the fields $\{t, d_t, \hat{d}_t, p_t, w_t\}$ and further refine this new dataset by removing possible outliers using the commonly used techniques—$z$-score and isolation forest (Zhao et al. 2019). Eventually, we prepare a refined dataset with 613 samples for our predictive analysis. We implement the predictive analysis using the combination of R language 4.1 and Python 3.8.

*Congestion Effect Quantification*. We first quantify the congestion effect of the increasing parcel flow on the average processing time of the robotic system. We begin with a simple univariate linear model shown as follows:

$$p_t = a\hat{d}_t + b + \xi_t, \tag{1}$$

where $a$ is the congestion coefficient, $b$ is the intercept term, and $\xi_t$ is the noise term. Note that the average processing time may be affected by the parcel weight (the robot travel speed may decrease as the parcel weight increases) and the potential fixed time effect. The impact of the parcel weight is examined by the following multivariate linear model:

$$p_t = a_1\hat{d}_t + a_2 w_t + b + \xi_t, \tag{2}$$

where $w_t$ is the parcel weight and $a_2$ is the weight coefficient. Furthermore, the fixed time effect is captured by the following multivariate linear model:

$$p_t = a_1\hat{d}_t + a_2 w_t + b + \sum\theta_i x_{it} + \xi_t, \tag{3}$$

where $x_{it}$ is a dummy variable that indicates whether period $t$ is the $i$-th time window of a day. The regression results for Models (1)–(3) are shown in Table 1. Specifically, the intercept term of Model (1) is $b = 35.4$ (with a standard error of 1.149), the congestion coefficient is $a = 0.0026$ (with a standard error of 0.00023), and the $R$-square score of Model (1) is 0.18. This result indicates that the congestion effect of increasing the parcel flow is significant. Moreover, by comparing the results for Models (1)–(3), we observe that the average processing time of the robotic system is insensitive to the parcel weight and free of the fixed time effect. One reason for little impact of parcel weight is that all the parcels have standard weights that normally vary from 0.5 to 1 kg. Hence, we conclude that the parcel flow is the main factor that shapes the average processing time of the robotic system.

**Table 1 Summary of Linear Regression Results**

| | Linear model (1) | Linear model (2) | Linear model (3) |
|---|---|---|---|
| Intercept | 35.4*** | 29.3*** | 31.5*** |
| Flow | 0.0026*** | 0.0028*** | 0.0028*** |
| Weight | — | 8.28* | 7.82* |
| Time fixed effect | Not included | Not included | Included |
| $R^2$ | 0.1785 | 0.1864 | 0.2057 |

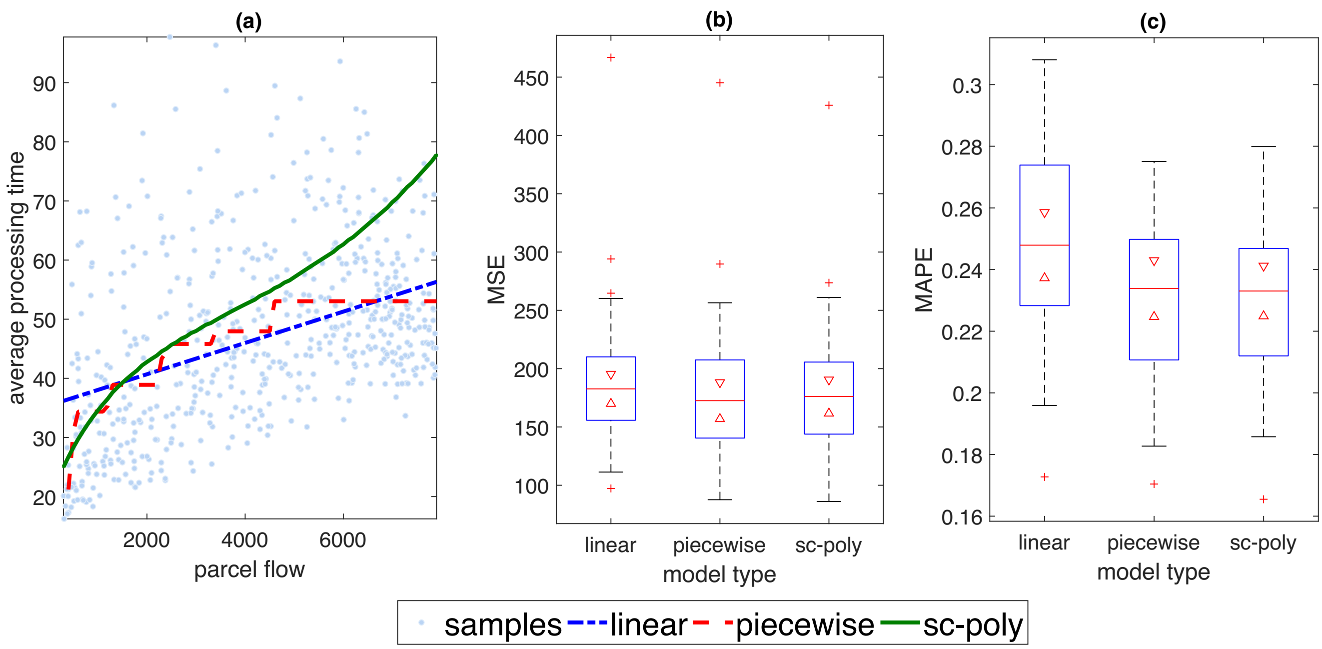
*$p < 0.05$, ***$p < 0.001$.

Next, we attempt to improve the predictive performance by developing generalized univariate regression models using $d_t$ as the input. First, we consider the nonparametric isotonic regression model that fits the data with an increasing and piecewise step function. In fact, similar piecewise step functions are often used in the literature (Zhang et al. 2009) to model workflow congestion in manufacturing/warehousing facilities. Applying the python toolkit, we obtain the fitted piecewise step function that is visualized in Figure 2a, and find that the $R$-square score of the isotonic regression is 0.27. Second, we take into account the polynomial regression model, which fits the data with the following polynomial function:

$$p_t = \sum_{i=1}^{n} a_i \hat{d}_t^{\alpha_i} + b + \xi_t, \tag{4}$$

where $n$ is the number of polynomial terms, $\alpha_i > 0$ is the order of the $i$-th polynomial term, and $a_i$ is the coefficient of the $i$-th polynomial term. Given that $\{n, \alpha_1, \ldots, \alpha_n\}$ are tuning parameters, we need to estimate the coefficients $\{a_1, \ldots, a_n, b\}$. Note that it is inappropriate to use the normal polynomial regression, because it does not impose any shape constraints and can output a function $f(d)$ that is non-monotonic in $d$.[2] To overcome this shortage, we use a machine learning approach, called shape-constrained polynomial regression (Curmei and Hall 2020), to obtain the polynomial function. The advantage of shape-constrained polynomial regression is that it enables us to incorporate prior knowledge on the shape of $p_r(d)$ (such as monotonicity) into the regression. In addition to the monotonicity, we impose two other shape constraints: $dp_r(d)$ is convex in $d$ and $p_r(300) \leq 25$, where the first constraint means that the total processing time of robotic systems convexly increases with parcel flow (which is intuitive to understand), and the second constraint comes from the manager's prior knowledge (i.e., when the parcel flow input is sufficiently smaller than 300, the average processing time will not exceed 25 seconds). The coefficients $\{a_1, \ldots, a_n, b\}$ can be estimated by solving the following least-squares minimization problem subject to shape constraints:

**Figure 2** Plot of Predictive Analysis for Congestion Effect: (a) Depicts the Linear, Piecewise, and Shape-Constrained Polynomial (labeled by "sc-ploy") Models for Congestion Effect of Increasing Parcel Flow on the Average Processing Time, and (b) and (c) Shows the MSEs and MAPEs of Different Models over the Validation Experiments [Color figure can be viewed at wileyonlinelibrary.com]




$$\min_{a_1,\ldots,a_n,b} \sum_{i=1}^{T} \left( \sum_{i=1}^{n} a_i \hat{d}_t^{\alpha_i} + b - p_t \right)^2,$$

s.t.,

$$\sum_{i=1}^{n} a_i \alpha_i \hat{d}_t^{\alpha_i - 1} \geq 0, \ \forall t, \quad \text{(monotonicity constraint)}$$

$$\sum_{i=1}^{n} a_i \alpha_i (\alpha_i + 1) \hat{d}_t^{\alpha_i - 1} \geq 0, \ \forall t, \quad \text{(convexity constraint)}$$

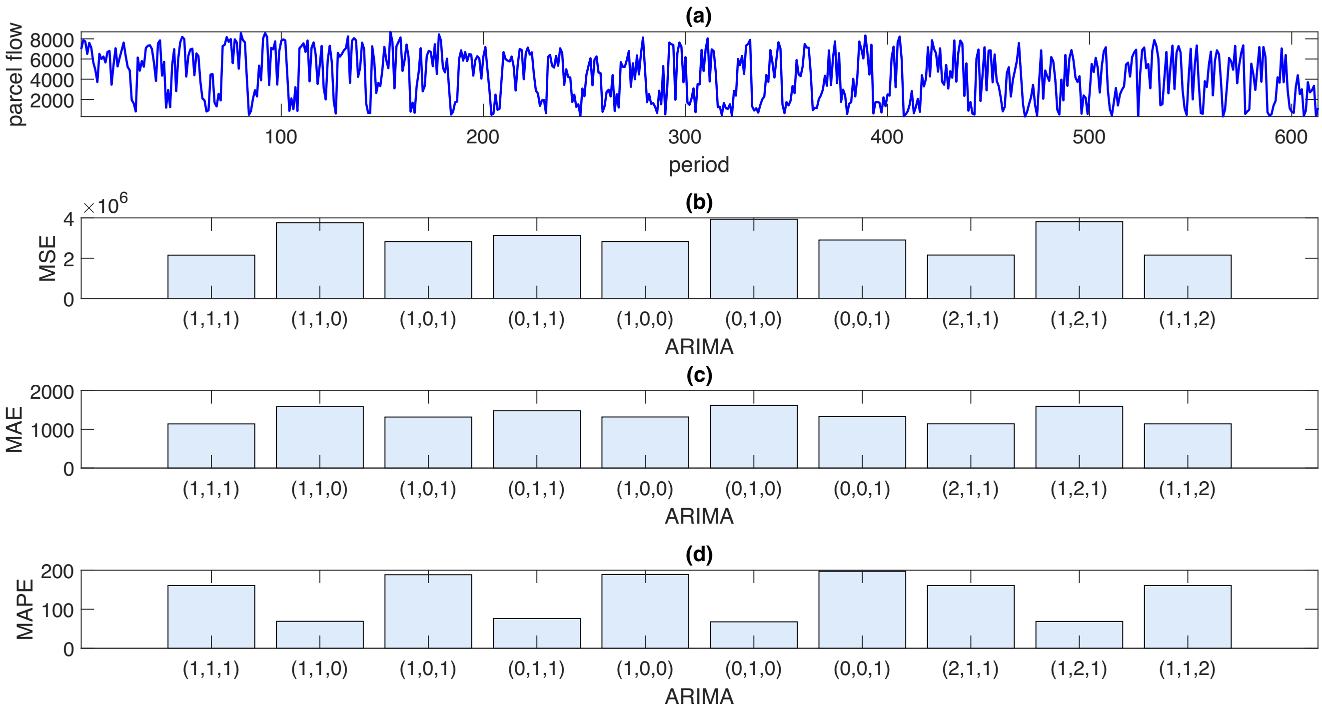
$$\sum_{i=1}^{n} a_i 300^{\alpha_i} + b \leq 25, \quad \text{(boundary constraint)}$$

which is a convex quadratic program that can be solved accurately and efficiently. By letting $n = 4$, $\alpha_1 = 1$, $\alpha_2 = 1.5$, $\alpha_3 = 2$, and $\alpha_4 = 3$, we obtain the result for the shape-constrained polynomial regression is $a_1 = 0.031$, $a_2 = -0.00058$, $a_3 = 3.578 \times 10^{-6}$, $a_4 = 0$, $b = 18.37$.

Based on the abovementioned discussion, we now have three candidate models for $p_r(d)$, which are shown in Figure 2a. Cross-validation experiments are conducted to compare the predictive performances of these models. In each experiment, we randomly select 95% of the samples as the training set and left the remaining 5% of the samples as the testing set. The prediction accuracy is measured by two commonly used metrics: mean square error (MSE) and mean absolute percentage error (MAPE). We repeat these experiments 50 times, and a summary of the predictive performances is shown in Figure 2b and c. The average MSEs of the linear, piecewise, and shape-constrained polynomial models over the 50 experiments are 192.07, 179.17, and 179.66, respectively, and the average MAPEs are 25%, 23%, and 23%, respectively. These results indicate that the linear model has the lowest accuracy, while the piecewise model has the highest accuracy, and the shape-constrained polynomial model is extremely close to the piecewise model.

*Parcel Flow Behavior*. The parcel flow time series is shown in Figure 3a. By observing the time series, we find that the parcel flow series is a nonstationary process with peak and off-peak patterns: the peak often arrives at the middle of a day (around 8 a.m.), and the off-peak arrives at the end of a night (around 12 p.m.). This observation is further confirmed by the augmented Dickey–Fuller test (the $p$-value of the test is 0.265 and rejects the stationary hypothesis). Following the standard technique of handling nonstationary series, we implement the logarithm transformation $y_t = \ln(d_t)$ to suppress the potential trends, and consider the transformed series $\{y_t\}$. A general ARIMA($p$, $i$, $q$) model is applied to fit the transformed series $\{y_t\}$, and the parameters ($p$, $i$, $q$) are selected following the rule of minimizing AIC and BIC. After fitting the model, we find that an ARIMA(1,1,1)-without-drift model can best fit the series $\{y_t\}$. Specifically, by defining $\Delta_t := y_t - y_{t-1}$ as the difference in $y_t$, the ARIMA (1,1,1) is given by the following:

**Figure 3 (a) Plot of Parcel Flow Time Series: Historical Parcel Flow Time Series and Comparisons of Out-of-Sample Performances of Different log-ARIMA(p,i,q) Models in Terms of (b) MSE, (c) MAE, and (d) MAPE, Respectively [Color figure can be viewed at wileyonlinelibrary.com]**




$$\Delta_t = \phi\Delta_{t-1} + \theta\varepsilon_{t-1} + \varepsilon_t, \tag{5}$$

where $\phi = 0.4906$ (with a standard error of 0.0358) is the autocorrelation coefficient, $\theta = -0.99$ (with a standard error of 0.005) is the moving average coefficient, and $\varepsilon_t$ is the white noise with a mean of zero and estimated variance of 0.4746. These results imply that the underlying parcel process is nonstationary and may follow a log-ARIMA(1,1,1)-without-drift model.

We also validate the effectiveness of the ARIMA (1,1,1) model in predicting $\{y_t\}$ by introducing nine other time series models (such as ARIMA(1,1,0), ARIMA(1,0,1), and ARIMA(2,1,1)) as benchmarks. Specifically, we use the samples from periods 1 to 600 as the training set and use the last 13 samples from periods 601 to 613 as the test set. We present the out-of-sample performances of these time series models, represented by MSE, MAE, and MAPE, in Figure 3b–d. Evidently, the ARIMA(1,1,1) model provides superior and robust predictive performances compared with other time series models.

The primary implications of the above discussion are as follows. First, the linear function, piecewise step function, and shape-constrained polynomial function are three candidate choices for modeling the average processing time of the robotic system and capturing the congestion effect. Although the piecewise function is the best fit for the data. It may not be the best modeling choice because its discontinuity and non-smoothness can lead to extra analytical difficulties. Conversely, the shape-constrained polynomial function not only has comparable predictive performance as the piecewise function, but also has good structural properties (like, differentiability). Hence, we choose to use the shape-constrained polynomial function to model the average processing time of the robotic system. Second, the underlying parcel flow process is verified to be nonstationary, highly uncertain, and less predictable. For the nonstationary parcel flow process, we need to develop effective approaches that are able to dynamically adjust the manual capacity based on evolving parcel flow information, such as online optimization (Li et al. 2021, Mokhtari et al. 2016) and MDP approach (Shi et al. 2019). Our theoretical analysis focuses on developing online algorithms for adjusting the manual capacity and exploring their distinctive features relative to the MDP approach (see section 5.2).

## 4. Theoretical Analysis

This section presents a theoretical analysis for the IoT-enabled human–robot hybrid sortation system. Section 4.1 formally describes our model, and section 4.2 provides the analytical results and algorithms for

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optimizing the parcel flow allocation and the manual capacity adjustment decisions.

### 4.1. The Model

Consider a human–robot hybrid sortation system operated over a planning horizon of $T$ periods, during which a manager monitors the average processing time of the hybrid system, that is, the KPI of the sortation center. The hybrid system consists of two components: the robotic (sub-)system is always online and has fixed capacity, and the manual (sub-)system has a set of $K$ identical manual sortation units. Let $d_t \in [0, \bar{d}]$ denote the parcel flow arriving in period $t$, where $\bar{d}$ is a predetermined upper bound. When $d_t$ is small and does not exceed the threshold $\underline{d}(<\bar{d})$, the robotic system alone provides a small average processing time $p_\epsilon$ because the congestion effect can be negligible. In this case, the manager allocates all the parcels to the robotic system. However, when $d_t$ is large and exceeds the threshold $\underline{d}$, the congestion effect becomes significant such that the average processing time of the robotic system is $p_r(d) + \xi_r$, where $p_r(d)$ is a smooth and increasing function of $d$, and $\xi_r$ is a white noise with a mean of zero. Meanwhile, the expected total processing time $P_r(d) = dp_r(d)$ of the robotic system is convexly increasing. A special case of $p_r(d)$ satisfying the above-stated assumptions has a linear form: $p_r(d) = a_r d + b_r$, where $a_r > 0$ is the time coefficient, and $b_r$ is the constant.

Because decision making for the scenario $d_t \in [0, \underline{d}]$ is trivial, our model assumes $d_t \in (\underline{d}, \bar{d}]$ and focuses on the scenario whereby the manual system is used. Given the parcel flow $d$, the average processing time of every manual sortation unit is $p_m(d) + \xi_m$. Specifically, $p_m(d)$ has the linear form,[3] $p_m(d) = a_m d + b_m$, with the time coefficient $a_m(>0)$ and constant term $b_m(\geq 0)$, and $\xi_m$ is a white noise with a mean of zero. The expected total processing time of a manual sortation unit is $P_m(d) = dp_m(d) = a_m d^2 + b_m d$. We also impose the following constraint: $p_r(d_t) \leq p_m(d_t)$ for $d_t \in (\underline{d}, \bar{d}]$, which means that the robotic system has a superior performance under the same conditions. Let $\gamma \in [0, 1]$ be the proportion of parcels allocated to the robotic system, and the remaining $(1 - \gamma)$ is allocated to the manual system. The expected total processing time of the robotic system is $P_r = \gamma d_t \times p_r(\gamma d_t)$. Given the number $k_t \in (0, K]$ of online manual units, it is optimal to equally distribute the parcels $(1 - \lambda)d_t$ to every online manual unit (see the online appendix for a rigorous proof) such that the expected total processing time of the manual system is $P_m = (1-\gamma)d_t\left(a_m \frac{(1-\gamma)d_t}{k_t} + b_m\right)$. Consequently, the expected average processing time of the hybrid system is

$$\bar{p}_t(\gamma, k_t, d_t) = \frac{P_r + P_m}{d_t} = \gamma p_r(\gamma d_t) + a_m \frac{(1-\gamma)^2 d_t}{k_t} + (1-\gamma) b_m. \quad (6)$$

*Cost Structure*. The manager has the following cost structure in each period $t$. The average processing time $\bar{p}_t(\cdot)$ of the hybrid system incurs a penalty cost $\ell(\bar{p}_t(\cdot))$ every period. We assume that $\ell(\cdot)$ is smoothly and convexly increasing, and a simple example of such $\ell(\cdot)$ is $\ell(\bar{p}_t) = c_p \bar{p}_t$, where $c_p > 0$ is the cost coefficient. The operation and maintenance of the robotic system incur a fixed cost $c_r$ per period. As the fixed cost $c_r$ can be viewed as a sunk cost, we normalize it to zero for ease of exposition. Given the number $k_t$ of online manual sortation units, the manager pays a quadratic operating cost $c_m k_t^2$ every period. This assumption is motivated by the practice of the China Post sortation center, and means that the marginal operating cost of the manual system increases with the number of online sortation units. Furthermore, switching each manual sortation unit on will incur a fixed setup cost $c_o$, because doing so involves several effort-consuming procedures (e.g., equipment preparation, workforce scheduling, and site cleaning).

*Impact of IoT Technology*. Without IoT technology, the manager is not able to responsively adjust the manual capacity decisions $\{k_t\}$ over time, and has to plan them following the static approach (whose details are shown in section 5.1). While, supported by IoT technology, the manager is able to (a) adjust the manual capacity over periods by switching on/off some manual sortation units based on freshly updated parcel flow information, and (b) observe some parcel flows in advance of their actual arrivals, which are referred to as "IoT advance information." More specifically, the manager at the beginning of period $t$ can observe the parcel flows $\{d_t, d_{t+1}, \ldots, d_{t+w-1}\}$, and $w(\geq 1)$ is the horizon of IoT advance information. In the presence of IoT technology, the sequence of events at a period $t$ is as follows. First, the manager decides the number $k_t$ of the online manual units based on the available information. Second, after observing the parcel flow $d_t$, the manager allocates the parcels between the robotic and manual systems by determining the proportion $\gamma$. By defining $C_t := \ell(\bar{p}_t(\cdot)) + c_m k_t^2 + c_o(k_t - k_{t-1})^+$ as the single-period cost in period $t$, the manager aims to minimize the expected operating cost rate $\frac{\sum_{t=1}^{T}\mathbb{E}[C_t]}{T}$ over the horizon.

Some remarks regarding the model are provided as follows. First, our model formulates the parcel sorting problem as a sequential decision problem, based on the operations of the China Post sortation center. It differs from prior models that are built on scheduling or queuing theories and focus on the details of truck or workforce scheduling (e.g., Khir et al. 2021, Novoa et al. 2018). Second, a sufficient condition ensuring the flexible manual capacity adjustment is that the manual system has sufficient workforce supply and can hire part-time workers in an on-demand style. In fact, this condition makes sense when part-time workers are conveniently hired and hourly paid via a third-party online workforce platform. Third, in our model, there is no parcel carryover between two consecutive periods because all the parcels arriving at the beginning of period $t$ can be sorted out at the end of this period. This implicit assumption is often used in the literature (Jarrah et al. 2016), since truck arrivals can be carefully scheduled such that sufficient time is left for sorting out all the parcels delivered by every truck.

### 4.2. Theoretical Results: Algorithm and Performance Guarantee

We first introduce the following preliminary notations used in the following analysis: $\|\boldsymbol{x}\|$ denotes the Euclidean norm of a vector $\boldsymbol{x}$, and $\|x\| = |x| = \sqrt{x^2}$ when $x$ is a scalar; $\Pi(\cdot)$ represents the following projection operator: $\Pi(y) = \max\{0, \min\{K, y\}\}$; given a function $f(x)$, $\nabla_x f(\cdot)$ represents the gradient (subgradient) of $f(\cdot)$ with respect to $x$ if $f(\cdot)$ is differentiable (non-differentiable); $\mathcal{O}(T^\alpha)$ (where $\alpha \geq 0$) represents the set of functions of order $T^\alpha$; $\mathbb{I}\{\cdot\}$ represents the indicator function, and $\mathbb{I}\{X\} = 1$ if $X$ is true. The complete set of notations and their meanings is provided in the online appendix.

**4.2.1. Parcel Flow Allocation Decision.** First, we consider the parcel flow allocation problem. Given the parcel flow $d_t$ and manual capacity $k_t$, the manager aims to minimize the expected average processing time $\bar{p}_t(\gamma, k_t, d_t)$ of the hybrid system as follows:

$$\min_{\gamma \in [0,1]} \{\bar{p}_t(\gamma, k_t, d_t)\}$$
$$= \min_{\gamma \in [0,1]} \left\{ \gamma p_r(\gamma d_t) + a_m \frac{(1-\gamma)^2 d_t}{k_t} + (1-\gamma) b_m \right\}. \quad (7)$$

Under the assumption that $dp_r(d)$ is convex, we find that $\bar{p}_t(\gamma, k, d_t)$ is also convex in $\gamma$, which implies the uniqueness of the optimal allocation decision $\hat{\gamma} := \arg\min_{\gamma \in [01]}\{\bar{p}(\gamma, k, d_t)\}$ for problem (7). By defining $\hat{p}(k, d_t) := \min_{\gamma \in [01]}\{\bar{p}_t(\gamma, k, d_t)\}$, we have:

LEMMA 1. *For the parcel flow allocation problem* (7),

(i) $\hat{\gamma}(k)$ *is decreasing in* $k$, *and* $\hat{\gamma}(0) = 1$;
(ii) $\hat{p}(k, d_t)$ *is increasing in* $d_t$;
(iii) $\hat{p}(k, d_t)$ *is decreasing and convex in* $k$;
(iv) $\hat{p}(k, d_t)$ *is submodular if* $b_m \geq u$, *where* $u := \max_{d \in [\underline{d}, \bar{d}]}\{p_r(d) - dp_r'(d)\}$

PROOF. Proofs for all the analytical results are provided in the online appendix. □

Lemma 1 can be interpreted as follows. Part (i) means that the manager should allocate less parcels to the robotic system as the online manual capacity $k$ is increased to reduce the burden on the robotic system. Part (ii) states that the hybrid system needs more time on sorting out parcels as the inbound parcel flow $d_t$ increases. In Part (iii), the monotonicity of $\hat{p}(k, d_t)$ with respect to $k$ indicates the positive effect of increasing $k$ on reducing the average processing time of the hybrid system; however, the convexity of $\hat{p}(k, d_t)$ indicates the positive effect diminishes as $k$ increases. Note that it is nontrivial to prove the convexity because we do not assume that $\bar{p}_t(\gamma, k, d_t)$ is jointly convex in $(\gamma, k)$. In fact, the assumption that the average processing time $p_m(\cdot)$ of every manual sortation unit has a linear form ensures the convexity, even though $\bar{p}_t(\gamma, k, d_t)$ is not jointly convex. In addition, Part (iv) presents the sufficient condition for the submodularity of $\hat{p}(k, d_t)$, namely the intercept term $b_m$ in $p_m(\cdot)$ is larger than a constant $u$. The sufficient condition is mild because it holds for a large class of functions $p_r(\cdot)$, for example, when $p_r(d) = a_r d + b_r$ is a linear or $p_r(d) = a_r d^2 + b_r d$ is quadratic.

**4.2.2. Manual Capacity Decision: Online Optimization.** We proceed to the manual capacity adjustment problem. Even though the manual capacity decisions should be integers in practice, we consider their linear relaxations in the following analysis, and will show that the continuous capacity decisions can be transformed into high-quality integer ones using normal rounding procedure in section 5.3.

Given the capacity decisions $k_{t-1}$ and $k_t$ in periods $t-1$ and $t$, the total cost in period $t$ can be expressed as: $C(k_t, k_{t-1}, d_t) = f(k_t, d_t) + c_o(k_t - k_{t-1})^+$, where $c_o(k_t - k_{t-1})^+$ is the manual capacity setup cost, and $f(k, d)$ is a myopic cost function given by

$$f(k, d) := \ell(\hat{p}(k, d)) + c_m k^2. \quad (8)$$

The objective is to minimize the expected cost rate by sequentially optimizing capacity decisions over time as follows:

$$\min_{\{k_t \in [0,K]\}} \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[C(k_t, k_{t-1}, d_t)]$$
$$= \min_{\{k_t \in [0,K]\}} \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}\left[f(k_t, d_t) + c_o(k_t - k_{t-1})^+\right], \quad (9)$$

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**Algorithm 1** Variant Online Gradient Descent for Manual Capacity Optimization

**Parameters:** Constant learning rate $\eta$, step size $\lambda$, and coefficient $h$

**Output:** $\left(k_{1,1}^{VOGD}, \cdots, k_{T,T}^{VOGD}\right)$

**Initialization:** $k_{1-w,1}^{VOGD}$

**for** period $t = 2 - w, \cdots, T$ **do**

Step 1. Implement the decision $k_{t,t}^{VOGD}$ if $t \geq 1$

Step 2. Initialize capacity decision for period $t + w \leq T$

$$k_{t,t+w}^{\ddagger} := k_{t-1,t-1+w}^{VOGD} - \eta \nabla_k f\left(k_{t-1,t-1+w}^{VOGD}, d_{t-1+w}\right),$$
$$k_{t,t+w}^{VOGD} := (1-h) k_{t-1,t-1+w}^{VOGD} + h\Pi\left(k_{t,t+w}^{\ddagger}\right)$$

Step 3. Update the capacity decisions for periods $t, \cdots, t + w - 1$ in a backwards manner:

**for** $\tau = \min\{t + w - 1, T\}, \min\{t + w - 2, T\}, \cdots, \max\{t, 1\}$ **do**

$$k_{t,\tau}^{VOGD} := \Pi\left(k_{t-1,\tau}^{VOGD} - \lambda g_t\left(k_{t-2,\tau-1}^{VOGD}, k_{t-1,\tau}^{VOGD}, k_{t,\tau+1}^{VOGD}\right)\right).$$

**end for**

**end for**

where $k_0 \equiv 0$, and the expectation operator is taken with respect to possible uncertainties in $d_t$. Obviously, the capacity setup cost $c_o(k_t - k_{t-1})^+$ leads to the difficulty of solving problem (9), because this term couples the capacity decisions at different periods.

We aim to develop online algorithms for optimizing the manual capacity adjustment decisions based on the theory of online (convex) optimization (Li et al. 2021, Mokhtari et al. 2016, Shalev-Shwartz 2011, Zinkevich 2003). Let $\mathcal{A}$ denote such an online algorithm, and let $\boldsymbol{k}^{\mathcal{A}} := \{k_1^{\mathcal{A}}, \ldots, k_T^{\mathcal{A}}\}$ denote the capacity decisions generated by the algorithm over time. Following the literature (e.g., Chen et al. 2017, Li et al. 2021, Mokhtari et al. 2016), given a sample path $\boldsymbol{d} := \{d_1, \ldots, d_T\}$ of parcel flows that sequentially arrive over periods, we measure the performance metric of $\mathcal{A}$ using the following dynamic regret:

$$\mathrm{Reg}(\mathcal{A}|\boldsymbol{d}) := \sum_{t=1}^{T}\left[C\left(k_t^{\mathcal{A}}, k_{t-1}^{\mathcal{A}}, d_t\right)\right] - \sum_{t=1}^{T}\left[C\left(k_t^*, k_{t-1}^*, d_t\right)\right], \tag{10}$$

where $\boldsymbol{k}^* = \{k_1^*, \ldots, k_T^*\}$ is the optimal capacity decisions for problem (9) under the assumption that the sample path $\boldsymbol{d}$ of parcel flows is known in hindsight. Moreover, we assess the expected performance of $\mathcal{A}$ using the expected dynamic regret $\mathbb{E}[\mathrm{Reg}(\mathcal{A}|\boldsymbol{d})]$, where the expectation is taken with respect to the uncertainties in the sample path $\boldsymbol{d}$. In addition, conditioned on the sample path $\boldsymbol{d}$, we consider the myopic solution $k_t^M := \arg\min_{k \in [0,K]}\{f(k, d_t)\}$, and define the following:

$$\mathcal{L}_{\boldsymbol{d}}^M := \sum_{t=1}^{T}\left|k_t^M - k_{t-1}^M\right|, \tag{11}$$

which is the sum of the distances $|k_t^M - k_{t-1}^M|$ between any two consecutive myopic solutions. Noting that $k_t^M$ depends on $d_t$, we can use $\mathcal{L}_{\boldsymbol{d}}^M$ as a metric that measures the volatility of the parcel flows. For example, we have $\mathcal{L}_{\boldsymbol{d}}^M = 0$ in the extreme case with $d_1 = , \ldots, = d_T$, implying that the parcel flows are not volatile at all.

Online Gradient Descent (OGD) is a conventional online algorithm that is known to update sequential decisions using only the freshest gradient information over time (Mokhtari et al. 2016, Shalev-Shwartz 2011, Zinkevich 2003). We find that the OGD can be applied to optimize the manual capacity decisions (please see section 1 in the online appendix for the details of OGD). Noticing that the OGD does not make full use of IoT advance information, we pose a natural question: "Is it possible to enhance the conventional OGD by incorporating IoT advance information?".

This question motivates us to design a Variant-OGD (VOGD), the details of which are shown in Algorithm 1. We explain how the VOGD works by considering the example capacity decision $k_{t+w}$ for the period $t + w$. Let $k_{s,t}$ denote the capacity decision made in period $s$ for a future period $t(\geq s)$. First, VOGD follows the basic idea of OGD and initializes the capacity decision $k_{t,t+w}^{VOGD}$ in period $t$ (see Step 2 of Algorithm 1) as follows:

$$k^{\ddagger}_{t,t+w} = k^{VOGD}_{t-1,t+w-1} - \eta \nabla_k f(k^{VOGD}_{t-1,t+w-1},\ d_{t+w-1}),$$
$$k^{VOGD}_{t,t+w} := (1-h)k^{VOGD}_{t-1,t+w-1} + h\Pi(k^{\ddagger}_{t,t+w}),$$

where $k^{\ddagger}_{t,t+w}$ is the target capacity level obtained using the gradient $\nabla_K f(k^{VOGD}_{t-1,t+w-1},\ d_{t+w-1})$. Second, recalling that IoT advance information refers to the observation of parcel flows $\{d_t, d_{t+1}, \ldots, d_{t+w-1}\}$ at the beginning of period $t$. That is, the myopic functions $\{f(k, d_t), f(k, d_{t+1}), \ldots, f(k, d_{t+w-1})\}$ and their gradients become advance knowledge to the manager. With this knowledge, the VOGD continues to update the capacity decision $k^{VOGD}_{\tau,t+w}$ ($\tau \in \{t+1, \ldots, t+w\}$) $w$ iterations from periods $t+1$ to $t+w$ (see Step 3 of Algorithm 1). In particular, the continuous updates of the capacity decision $k^{VOGD}_{\tau,t+w}$ follow the classical rule of iterative searching a global optimizer (Boyd and Vandenberghe 2004) for the convex function, $f(k_t, d_t) + c_o(k_t - k_{t-1})^+ + c_o(k_{t+1} - k_t)^+$, with a step size $\lambda$ according to its sub-gradient:

$$g_t(k_{t-1}, k_t, k_{t+1}) = \nabla_K f(k_t, d_t) + c_o(\mathbb{I}\{k_t > k_{t-1}\} - \mathbb{I}\{k_{t+1} > k_t\}). \quad (12)$$

By letting $G$ be a global upper bound of $\|\nabla_K f(\cdot)\|$ (Lemma 2 in the online appendix proves the existence of $G$), we have:

Proposition 1. *For $w \geq 1$, given a sample path $\boldsymbol{d}$ of parcel flows, by setting $(\eta, h, \lambda)$ to satisfy $\frac{1}{2c_m} > \eta h$ and $\lambda = \frac{K}{G\sqrt{w}}$, we have $\mathrm{Reg}(VOGD|\boldsymbol{d}) \leq \mathcal{O}(\mathcal{L}^M_{\boldsymbol{d}}/\sqrt{w})$.*

Proposition 1 shows that the regret of the VOGD over a sample path $\boldsymbol{d}$ of parcel flows is upper bounded by a term $\mathcal{O}(\mathcal{L}^M_{\boldsymbol{d}}/\sqrt{w})$ that decreases as the horizon $w$ of IoT advance information increases at the rate of $\mathcal{O}(1/\sqrt{w})$. This result partially demonstrates the value of IoT technology for the human–robots hybrid sortation systems. Notably, $1/\sqrt{w}$ is also convex in $w$, and the convexity implies a diminishing return of increasing $w$ on improving the value of IoT technology. When investing in IoT technology, parameter $w$ is an important strategic choice for decision-makers, because an expansion in $w$ consumes costly efforts (e.g., more IoT devices and tighter coordination are needed). The diminishing return reminds decision-makers to carefully choose $w$, by considering that a large $w$ may not generate a sufficient surplus to cover the growing investment. This insight is further confirmed by the numerical results in section 5.3. Moreover, the inequality $\frac{1}{2c_m} > \eta h$ also illustrates how to choose the tuning parameters $h$ and $\eta$, by claiming that their product is within the upper bound $\frac{1}{2c_m}$.

Our results are similar to those developed by Li et al. (2021), however, have the following distinctive features. First, the work of Li et al. (2021) is based on the assumption that the myopic function $f(k, d_t)$ is not only strongly convex but also sufficiently smooth (called "$l$-smooth"); unlike their work, we only require the assumption of strong convexity. Second, as a result of relaxing the smoothness assumption, our online algorithm also differs from theirs by incorporating a new coefficient $h$ and initializing the capacity decision using a weighted combination of the previous and newly updated capacity decisions (see Step 2 of Algorithm 1). Third, our regret bound decays with $w$ at the rate of $\mathcal{O}(1/\sqrt{w})$, which is slower than the exponential rate developed by Li et al. (2021), without the smoothness assumption.

Proposition 2. *When the underlying parcel process follows a log-ARIMA(1,1,1)-without-drift process and $b_m \geq u$, the long-term expected regret rate of the VOGD is zero (i.e., $\lim_{T\to\infty} \frac{\mathbb{E}[\mathrm{Reg}(VOGD|\boldsymbol{d})]}{T} = 0$).*

Proposition 2 shows that the long-term expected regret rate of the VOGD is zero under mild conditions, namely the underlying parcel process follows a log-ARIMA(1,1,1)-without-drift process (being verified in the predictive analysis), and the intercept term $b_m$ in the average processing time of every manual sortation unit is larger than the threshold $u$ (see Lemma 1(iv)). This result provides a statistical guarantee for the effectiveness of our algorithm by demonstrating that the VOGD has no regret in an average sense as the planning horizon increases. Moreover, the law of large numbers implies that by sampling a large number $N$ of different long sample paths (i.e., $T$ is a large number) of parcel flows, we have

$$\frac{\mathbb{E}\left[\mathrm{Reg}(VOGD|\boldsymbol{d})\right]}{T} \approx \frac{1}{N}\sum_{i=1}^{N} \frac{\mathrm{Reg}\left(VOGD|\boldsymbol{d}^{(i)}\right)}{T} = \frac{1}{N}\sum_{i=1}^{N}\left[R^{VOGD}_{\boldsymbol{d}^{(i)}} - R^*_{\boldsymbol{d}^{(i)}}\right] = 0,$$

where $\boldsymbol{d}^{(i)}$ is the $i$-th sample path, $R^{VOGD}_{\boldsymbol{d}^{(i)}} = \frac{\sum_{t=1}^{T} C(k^{VOGD}_t,\ k^{VOGD}_{t-1},\ d^{(i)}_t)}{T}$ is the cost rate of the VOGD, and $R^*_{\boldsymbol{d}^{(i)}} = \frac{\sum_{t=1}^{T} C(k^*_t,\ k^*_{t-1},\ d^{(i)}_t)}{T}$ is the optimal cost rate (in hindsight) over $\boldsymbol{d}^{(i)}$. In other words, if we repeat the application of our algorithm to numerous different long sample paths, the average performance loss of the VOGD (relative to the optimal cost rate in

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hindsight) can be very small or close to zero. This result is also verified by extensive simulation experiments, as shown in section 5.3.

## 5. Value of IoT-Enabled Online Optimization

This section assesses the value of IoT-enabled online optimization relative to two benchmark approaches. One benchmark is the static approach currently used by the China Post sortation center in the absence of IoT technology, and the other benchmark is the conventional MDP approach used for dynamically adjusting manual capacity in the presence of IoT technology. In sections 5.1 and 5.2, theoretical results are also derived to illustrate potential limitations of the two benchmark approaches, respectively. In section 5.3, through extensive simulation experiments, we compare the performances of our online algorithm and the two benchmark approaches, and develop a sensitivity analysis of their performances with respect to several key parameters.

### 5.1. Benchmark I-Static Approach

Under the static approach, the entire horizon is divided into a few subhorizons with $L$ periods in each subhorizon. At the beginning of the $i$-th subhorizon, the warehouse manager observes the historical parcel flows $\{d_1, \ldots, d_{(i-1)L}\}$ and plans the capacity decisions for this subhorizon by solving the following problem:

$$\min_{\substack{k_{(i-1)L+1} \in [0, K] \\ ,\ldots, \\ k_{iL} \in [0, K]}} \frac{1}{L} \sum_{j=(i-1)L+1}^{iL} \times \left\{ \mathbb{E}\left[ f(k_j, d_j) | d_1, \ldots, d_{(i-1)L} \right] + c_o (k_j - k_{j-1})^+ \right\}, \quad (13)$$

where the expectation in problem (13) is conditioned on the historical flow information. Based on the convexity of $f(\cdot)$, we know that problem (13) is a convex nonlinear program, whose optimal capacity decisions $\boldsymbol{k}_i^S := \{k_{(i-1)L+1}^S, \ldots, k_{iL}^S\}$ for (13) can be obtained using the classical gradient-based search method.

Some remarks on the implementation of the static approach are as follows. At the beginning of the $i$-th subhorizon, the warehouse manager first updates parcel flow information and then calculates the capacity decisions $\boldsymbol{k}_i^S$ by solving problem (13) based on the fresh parcel flow information. During the $i$-th subhorizon, the warehouse manager stops updating parcel flow information and strictly executes the capacity decisions $\boldsymbol{k}_i^S$ for every period of the subhorizon. The abovementioned steps are repeated in a rolling horizon way. Apparently, the key tuning parameter when implementing the static approach is $L$. On the one hand, the efficiency of the static approach increases with increasing $L$ by saving efforts needed to adjust the capacity decisions. On the other hand, the responsiveness of the static approach to nonstationary parcel flows reduces as $L$ increases by decreasing the frequency of information updates. The following result shows the impact of changing $L$ on the capacity decisions.

PROPOSITION 3. *When the parcel flows follow an log-ARIMA*(1,1,1)*-without-drift process, and the robotic system has a linear-form average processing time, the optimal capacity decisions for* (13) *converge to K as L increases to be large* ($L \to \infty$).

Proposition 3 reveals a major limitation of the static approach—its capacity decisions for every subhorizon will converge to the maximum capacity level $K$ as the length $L$ of the subhorizon increases to a large value. The reason for this result is as follows. When the parcel flows follow the nonstationary log-ARIMA(1,1,1)-without-drift process, the variance of random parcel flow $d_j$ increases with $j \in [(i-1)L+1, iL]$, and the long-term parcel flow $d_\infty$ has an unbounded variance as $L$ becomes large (see Lemma 1 in the online appendix). In this case, all the manual capacity will be exhausted to hedge against the increasingly huge risks in every subhorizon. Although Proposition 3 only demonstrates the convergence of the capacity decisions when the processing time of the robotic system is linear, some numerical results show that the convergence behavior continues to hold even when it is nonlinear (see Figure 1 in the online appendix). More importantly, the convergence speed is very fast, that is, the manual capacity decisions can quickly reach the maximum capacity level even when the value of $L$ is moderate (e.g., $L = 5$). In the practice of China Post sortation center, parameter $L$ is often set to be moderate or large (e.g., $L = 10$) for efficiency. Such a choice of $L$ partially explains our observation that the China Post sortation center often runs out of its manual capacity.

### 5.2. Benchmark II-MDP Approach

In this subsection, we present the details of the MDP approach for optimizing the manual capacity decisions in the presence of IoT technology. We let $\boldsymbol{s}_t = (k_{t-1}, d_t, \ldots, d_{t+w-1})$ denote the state space of the dynamic program at the beginning of period $t$, where $k_{t-1}$ is the manual capacity of period $t-1$, and $d_t, \ldots, d_{t+w-1}$ are IoT advance information. The state space evolves according to $\boldsymbol{s}_{t+1} = (k_t, d_{t+1}, \ldots, d_{t+w})$. By defining $V_t(\boldsymbol{s}_t)$ as the optimal value function, we express the dynamic program as follows:

$$V_t(k_{t-1}, d_t, \ldots, d_{t+w-1}) = \min_{k \in [0, K]} \{f(k, d_t) + c_o(k - k_{t-1})^+ + \mathbb{E}[V_{t+1}(k, d_{t+1}, \ldots, d_{t+w})]\}, \qquad (14)$$

with the ending condition $V_{T+1}(\cdot) \equiv 0$, where the expectation is taken with respect to the uncertainty in $d_{t+w}$. The following result shows the structural properties of $V_t(\cdot)$.

LEMMA 2. *For* $t = 1, \ldots, T$, (i) $V_t(\cdot)$ *is decreasing in* $k_{t-1}$ *while increasing in* $d_i$, $i \in \{t, \ldots, t+w-1\}$; (ii) $V_t(\cdot)$ *is convex in* $k_{t-1}$.

Lemma 2(ii) implies the convexity of the objective function $H_t(k) := f(k, d_t) + c_o(k - k_{t-1})^+ + \mathbb{E}[V_{t+1}(k, d_{t+1}, \ldots, d_{t+w})]$. By defining $\tilde{k}_t := \arg\min_k \{H_t(k)\}$ as the target capacity level, we obtain that the MDP approach outputs the capacity decision $k_t^{MDP}$ that is closest to $\tilde{k}_t$ as follows: if $0 \leq \tilde{k}_t \leq K$, then $k_t^{MDP} = \tilde{k}_t$; if $\tilde{k}_t < 0$, $k_t^{MDP} = 0$; if $\tilde{k}_t > K$, $k_t^{MDP} = K$. The abovementioned discussion shows that the computational efficiency of the MDP approach depends on how to search for the target capacity level $\tilde{k}_t$. The literature (e.g., Shi et al. 2019) has shown that the searching efficiency for $\tilde{k}_t$ can be improved by exploiting the possible monotonicity $\tilde{k}_t$ with respect to the state space $\boldsymbol{s}_t$.

PROPOSITION 4. *For* $t = 1, \ldots, T$, (i) $\tilde{k}_t$ *is increasing* $k_{t-1}$; (ii) $\tilde{k}_t$ *can be non-monotonic in* $d_i$, $i \in \{t, \ldots, t+w-1\}$.

Proposition 4(i) confirms the monotonicity of the target capacity level $k_t^{MDP}$ with respect to the first component $k_{t-1}$ of the state space $\boldsymbol{s}_t$. Unfortunately, as shown by Proposition 4(ii), monotonicity does not necessarily hold for the remaining components $d_t, \ldots, d_{t+w-1}$ of the state space (see Figure 2 in the online appendix for an example). The reason for this non-monotonicity is as follows. When the parcel flow is small or moderate, the congestion effect of the robotic system is moderate, and the fraction of the parcels allocated to the robotic system increases with the parcel flow (i.e., the parcel flow allocation decision increases with the parcel flows); accordingly, less manual capacity is required. When the parcel flow is large, the congestion effect is strong, and the fraction of the parcels allocated to the robotic system decreases as the parcel flow increases; thus, more manual capacity is required. It should be noted that the monotonicity of $\tilde{k}_t$ with respect to $k_{t-1}$ is not quite useful for reducing the search space because it takes effect only when the remaining components $\{d_t, \ldots, d_{t+w-1}\}$ of the state space are the same. Therefore, we conclude that the computational efficiency of the MDP approach can be low.

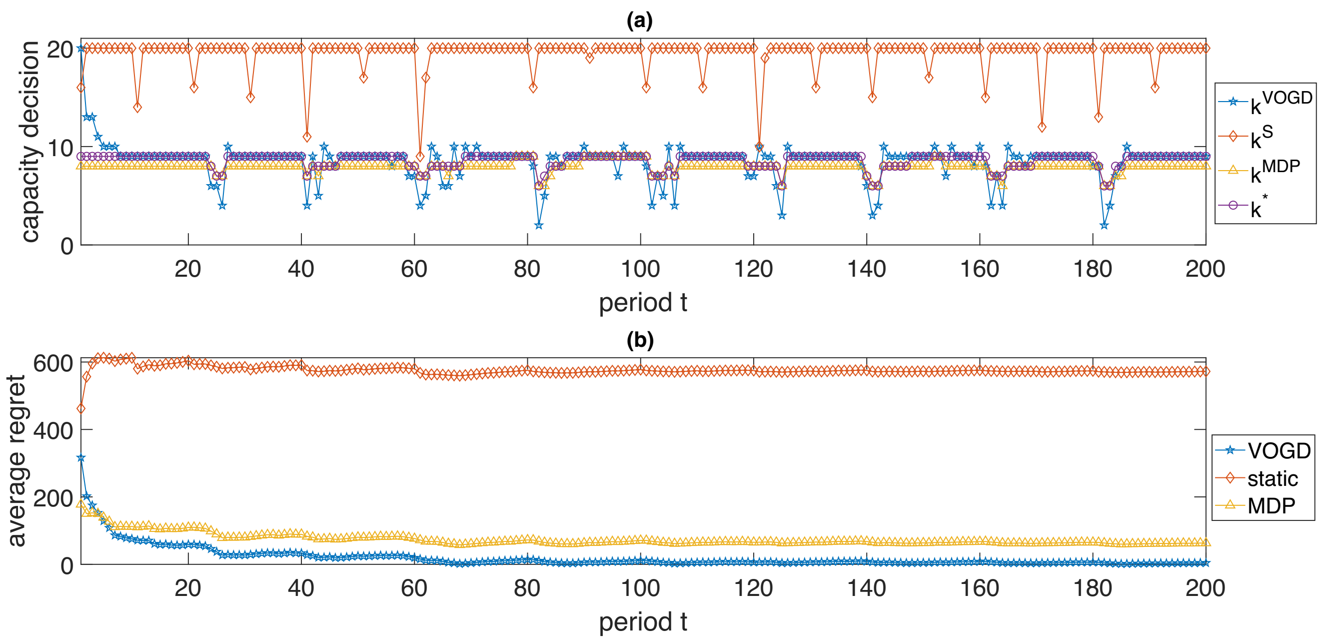
We now compare the online optimization and the MDP approach. First, the computational efficiency of the online algorithm is high, because it only uses the freshest gradient information to adjust the manual capacity decisions. Conversely, the computational efficiency of solving the high-dimensional MDP can be low without knowing useful structural properties that can be used to reduce the search space. Thus, we say that the online algorithm is more compatible with IoT technology, which requires higher computational efficiency than the MDP approach. Second, the MDP approach requires prior distributional knowledge about the underlying parcel flow process. Unlike the MDP approach, the online algorithm deals with a more challenging version of the manual capacity problem, without requiring prior distributional knowledge about the underlying parcel flow process. Third, to the best of our knowledge, it is difficult to develop a theoretical guarantee for the effectiveness of the MDP approach. Instead, Propositions 1 and 2 demonstrate the effectiveness of the online algorithm by providing theoretical bounds on the regret.

### 5.3. Simulation Result

This part complements the theoretical findings via extensive simulation experiments, based on the real-life dataset provided by the China Post sortation center. We try to provide numerical insights into the following questions: (a) How can we evaluate the effectiveness of the online optimization and benchmark approaches? (b) How can we assess the value of the China Post sortation center's IoT program? (c) What is the impact of key parameters on the value of the IoT program?

The setup of the simulation experiments is as follows. The average processing time of the robotic system follows the polynomial function shown in section 3: $p_r(d) = 18.37 + 0.031d - 0.00058d^{3/2} + 3.578 \times 10^{-6}d^2$, and the average processing time of every manual sortation unit follows a linear function $p_m(d) = 0.08d$. It is easy to verify that these two functions satisfy the constraint $p_m(d) \geq p_r(d)$. As the sortation center has 20 manual sortation units in total, we set the maximum capacity level $K$ to be 20. Under the current IoT program, the horizon $w$ of IoT advance information is 1. In addition, the underlying parcel flow process follows the log-ARIMA

**Figure 4 Plot of Effectiveness of Different Approaches for a Realistic Sample Path of 200 Periods: (a) Depicts the Capacity Decisions and (b) Depicts the Average Costs over Time [Color figure can be viewed at wileyonlinelibrary.com]**




(1,1,1)-without-drift process (with $\phi = 0.49$, $\theta = -0.99$, $\sigma^2 = 0.475$) shown in section 3. This is also the prior knowledge when using the static and MDP approaches. Without specific financial datasets (for confidentiality reasons), we follow the rough estimates provided by the sortation center and simulate the cost parameters by assigning values as follows: $c_m$ is normalized to 1; the penalty cost for the average processing time has a linear form $\ell(\bar{p}) = c_p\bar{p}$; the cost parameters $c_o$ and $c_p$ are set to be 2 and 9, respectively.

We repeat the abovementioned experiment $N = 20$ times, and the $i$-th ($i \in \{1, \ldots, N\}$) experiment consists of the following steps. First, we sample a path of the parcel flows from the log-ARIMA(1,1,1)-without-drift process, denoted as $\boldsymbol{d}^{(i)} = \{d_1^{(i)}, \ldots, d_T^{(i)}\}$, where $T = 200$. Second, we simulate the ideal scenario wherein all the parcel flows are known in hindsight, and compute the optimal capacity decisions $\boldsymbol{k}^* = \{k_1, \ldots, k_T\}$ for the ideal scenario. As we are interested in integer solutions, we employ a deterministic dynamic program to search $\boldsymbol{k}^*$ within $\mathcal{O}(T^2)$ times. Third, we simulate a realistic scenario in which the parcel flows are sequentially revealed over periods, and use the static approach (with $L = 10$), the MDP approach, and the VOGD (with $\lambda = 0.5$, $\eta = 0.1$, $h = 0.5$) to generate the corresponding capacity decisions $\boldsymbol{k}^S$, $\boldsymbol{k}^{MDP}$, and $\boldsymbol{k}^{VOGD}$, respectively. In particular, we implement the MDP approach in a rolling horizon way, and increase the efficiency of computing $\boldsymbol{k}^{MDP}$ using a common heuristic that only looks three periods ahead (Heyman and Sobel 2004, Shi et al. 2019). After the abovementioned steps, we can get four groups of capacity decisions for the sample path $\boldsymbol{d}^{(i)}$, based on which the corresponding realized costs can be computed.

**Table 2 Performance Comparison of Different Approaches for a Realistic Problem Instance with 200 Periods**

| | Computation times (seconds) | Performance gap (%) | Performance improvement (%) |
|---|---|---|---|
| Online optimization | 0.47 | 1.18 | 150.54 |
| Static planning approach | 979.88 | 153.51 | — |
| Markov dynamic program approach | 3489.72 | 16.95 | 116.77 |

For the $i$-th experiment ($i = 1, \ldots, N$), we evaluate the effectiveness of the online algorithm and two benchmark approaches (relative to the optimal cost rate in hindsight) using the performance gap, $\frac{R_Z^{(i)} - R_*^{(i)}}{R_*^{(i)}} \times 100\%$, where $R_Z^{(i)}$ is the cost rate under the approach $Z \in \{VOGD, S, MDP\}$, and $R_*^{(i)}$ is the optimal cost rate in hindsight. We also assess the value of the IoT program, $\frac{R_S^{(i)} - R_{VOGD}^{(i)}}{R_{VOGD}^{(i)}} \times 100\%$, via the average cost saving generated by our online algorithm relative to the static approach.

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**Figure 5** Plot of the Impact of Changing Key Parameters ($w$, $K$, $c_o$, $c_m$) on the Performance Gap of the VOGD Relative to the Optimal Cost Rate in Hindsight; Given a Key Parameter, the Boxplot is a Summary of the Gaps Over $N = 20$ Experiments [Color figure can be viewed at wileyonlinelibrary.com]

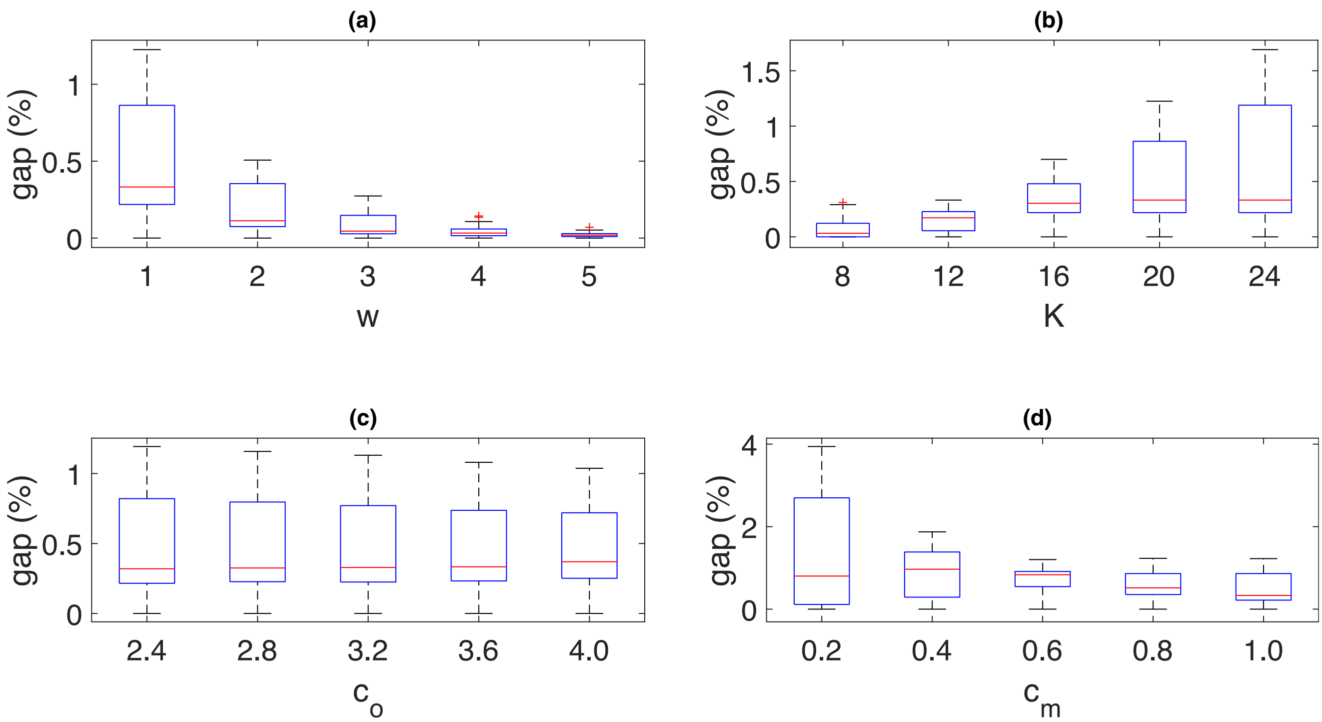
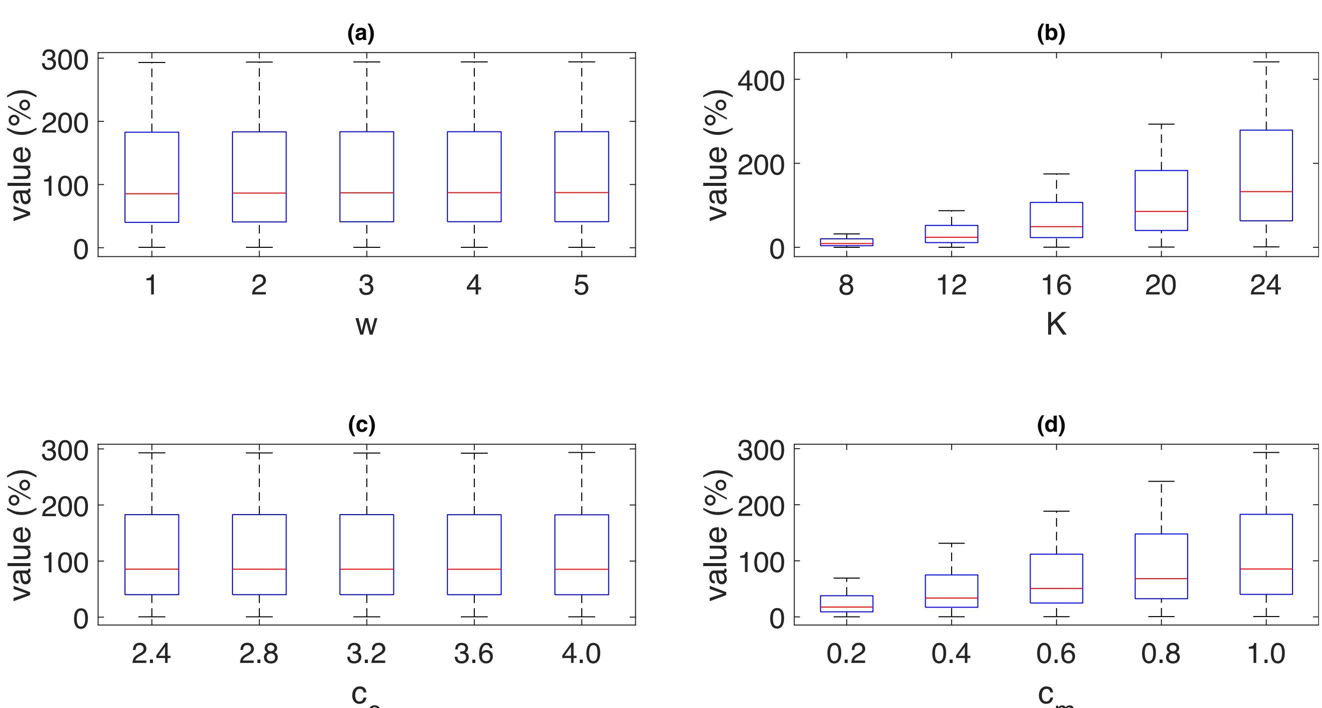



**Figure 6** Plot of the Impact of Changing Key Parameters ($w$, $K$, $c_o$, $c_m$) on the Average Value of the IoT Program Relative to the Static Approach; Given a Key Parameter, the Boxplot is a Summary of the Values over $N = 20$ Experiments [Color figure can be viewed at wileyonlinelibrary.com]



We first report on the effectiveness of the VOGD and two benchmark approaches. Specifically, Figure 4 depicts the effectiveness of these approaches for a sample path of 200 periods. From Figure 4a, we find that the capacity decisions $\boldsymbol{k}^S$ generated by the static approach often converge to the maximum capacity level $K = 20$, which is consistent with the findings shown in Proposition 1. However, the optimal capacity decisions $\boldsymbol{k}^*$ do not exceed half of the maximum manual capacity, implying that the static approach results in a significant waste of manual capacity. Unlike the static approach, the VOGD and MDP approaches generate the decisions $\boldsymbol{k}^{VOGD}$ and $\boldsymbol{k}^{MDP}$ that are close to the optimal ones, and hence can lead to better performances.

We also visualize how the average regret under different approaches evolve over time in 4(b), which is defined as $\frac{\sum_{i=1}^{t}[C(k_i^Z,\ k_{i-1}^Z,\ d_i)-C(k_i^*,\ k_{i-1}^*,\ d_i)]}{t}$, $Z \in \{S,\ MDP,\ VOGD\}$. Evidently, the average costs under the capacity decisions $\boldsymbol{k}^S$, $\boldsymbol{k}^{MDP}$, and $\boldsymbol{k}^{VOGD}$ all decrease and converge as $t$ increases, and $\boldsymbol{k}^{VOGD}$ leads to the largest decrease in cost and the fastest convergence. A summary of the effectiveness of the approaches is presented in Table 2, which reports the computational times, performance gap, and cost performance improvement (relative to the static approach). Apparently, the VOGD has the highest computational efficiency (no longer than 0.5 second), provides the smallest performance gap (approximately 1%), and has the largest performance improvement (approximately 150%). The abovementioned results demonstrate the superior effectiveness of the VOGD. Additionally, we conclude that if the IoT-driven online optimization is adopted, the IoT program can generate a 150% cost reduction relative to the China Post sortation center's current policy in use.

Next, we investigate the impact of key parameters on the effectiveness of our online algorithm. Figure 5 shows the average performance gaps of the VOGD (relative to the optimal cost rate in hindsight), and indicates that the VOGD provides robust and high-quality solutions (the maximum performance gap does not exceed 1.5%) regardless of the changes of the parameters. More specifically, by observing Figure 5a, we find that, as the horizon $w$ of the advance IoT information increases, the performance gap quickly shrinks for $w \in [1, 2]$ and then slowly decreases for $w \in [3, 5]$. This finding is consistent with the result in Proposition 1 and confirms the diminishing return of increasing $w$. In addition, we also find that the variation of the performance gaps quickly decreases as $w$ increases, which implies that IoT advance information helps stabilize the performance of our online algorithm. Meanwhile, the average performance gap of the MDP approach is 20.13%, and the average computation time is 3330 seconds. Such a long computation time makes the MDP approach less adaptable to IoT technology, because the optimization in IoT architecture has to be completed within very short time frames (within seconds).

Figure 5b shows that the average performance gap gradually increases as the maximal manual capacity $K$ increases. The reason for the performance loss is as follows. As $K$ increases, the feasible set $\{k_t : 0 \le k \le K\}$ expands; unlike the sophisticated search-based algorithms, the VOGD does not extensively search for the best solutions in the expanding feasible set. Nevertheless, the average performance loss is considerably small and below 0.8%. Figure 5c and d show that the average performance gap is insensitive to the change of the cost parameter $c_o$, and decreases with the cost parameter $c_m$.

Figure 6 depicts the value of the IoT program as the key parameters change. Specifically, Figure 6a shows that the value of the IoT program is insensitive to $w$ because of the diminishing return of increasing $w$. This result also provides an important implication that the China Post sortation center does not have to expand the horizon of IoT advance information. Figure 6b shows that the value of the IoT program also increases with the maximal capacity $K$, by avoiding wasting the manual capacity. Figures 6c and d indicate that the value is insensitive to the change in the cost parameter $c_o$ and increases in the cost parameter $c_m$. As a summary of the abovementioned experiments with different parameters, the overall average performance gap is 0.22%, and the overall average value is 97.86%.

In addition, we also compare the performances of the VOGD and OGD (see Figure 3 in the online appendix), and find that the former algorithm outperforms the latter in terms of responsiveness and robustness. This observation again highlights the positive role of IoT advance information on improving and stabilizing the performance of online algorithm.

## 6. Concluding Remarks

Although automation has been widely used in modern warehouses, human workers are not completely replaced by robots. Many warehouses employ human–robot hybrid systems to handle warehousing tasks and face the problems on how to coordinate manual and robotic systems to reduce operating costs. Based on the practice of a China Post sortation center, we present an analytics study on how to improve the sorting efficiency of a human–robot hybrid system via IoT-driven online optimization.

We start with a predictive analysis of a dataset provided by the China Post sortation center, which helps understand the congestion effect of increasing parcel

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flow on the performance of robotic system and the behavior of the parcel flow process. Based on the predictive findings, we develop a theoretical model for analyzing the problem and design an online algorithm that integrates IoT advance information to dynamically adjust the manual capacity decisions. The extensive simulation experiments indicate that our online algorithm outperforms the China Post sortation center's current policy and the conventional MDP approach, and hence could lead to considerable operating cost reduction.

This study can be extended to different directions. First, our model assumes that all manual sortation units are identical and that the average processing time of each unit increases linearly with the parcel flow. An interesting extension is to examine non-identical manual sortation units with nonlinear average processing times. Second, this study considers the problem of improving the parcel sorting efficiency from the perspective of manual capacity optimization. While, the sorting efficiency can be dependent on truck arrival/departure scheduling. Another interesting topic is on how to jointly optimize the manual capacity and truck arrival/departure scheduling in an IoT context.

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### Notes

[1]Someone may ask whether it is possible to expand the capacity of the robotic system by adding more robots to the system? This intuitive idea is not favorable because more robots can intensify the congestion effect in a robotic system. In fact, the maximum number of robots in a robotic system is well designed and fixed in practice.

[2]For example, if we fit the data with a polynomial function of order 3, we have $p_r(d) = -11.26d^3 - 95.77d^2 + 156.55d + 47.03$, which is non-monotonic in $d$

[3]The linear form of $p_m(d)$ comes from the rough estimate about the average processing time of a manual sortation unit provided by the China Post sortation center.

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## Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

**Appendix S1:** Online Gradient Descent.
**Appendix S2:** Proofs for Analytical Results.
**Appendix S3:** Additional Technical Results and Numerical Observations.

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